

QRC Usefulness Atlas: Visual Abstract

Outcome composition, family structure, anti-circularity robustness, and quantum-attribution guardrails in one reproducible view.

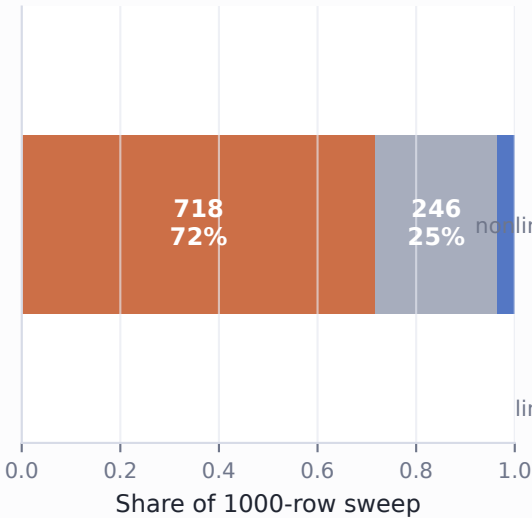
ATLAS ROWS
1,000
50 generators

QRC-USEFUL
36
3.6% of sweep

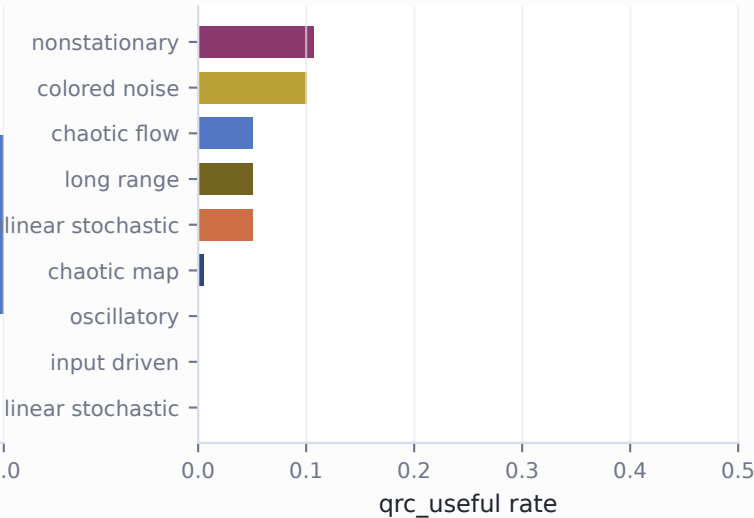
META-MODEL
R2 0.76
AUC 0.81

NO-PROXY CHECK
R2 0.76
AUC 0.81

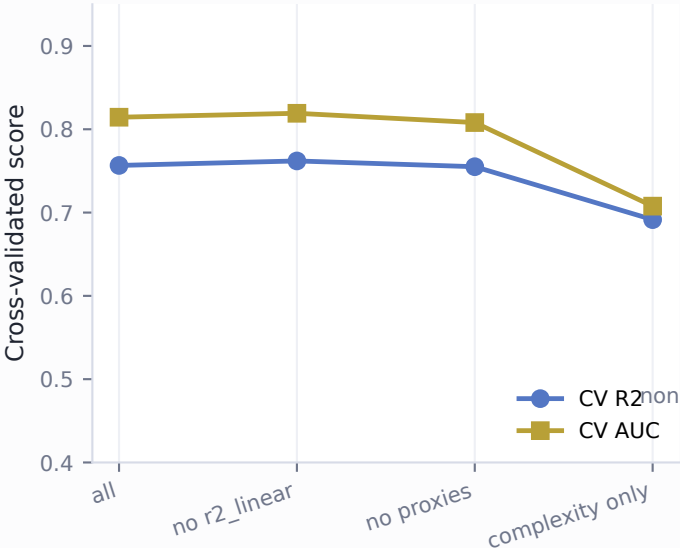
Outcome composition



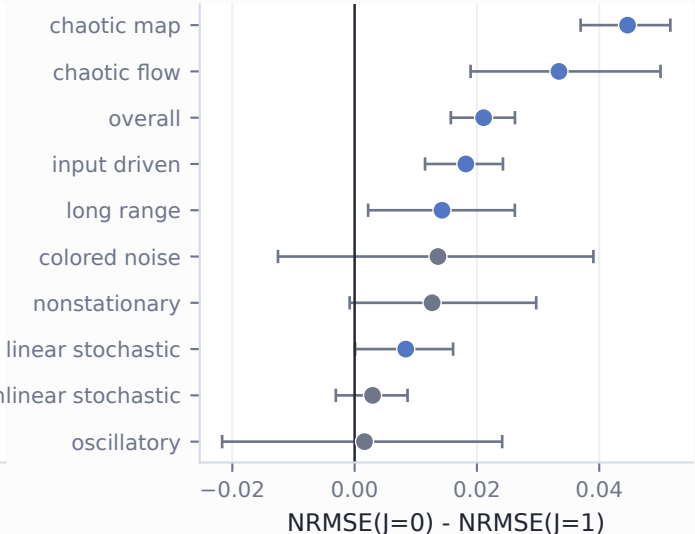
Family useful-rate ranking



Anti-circularity feature-set checks



Paired coupling-control intervals



Main claim: QRC usefulness is dataset-conditional and learnable from measured properties.

Strongest family: nonstationary: useful rate 10.7%.

Guardrail: Overall J=1 vs J=0 CI [0.016, 0.026] crosses zero.

Not claimed: No broad average QRC superiority and no coupling/entanglement mechanism claim.